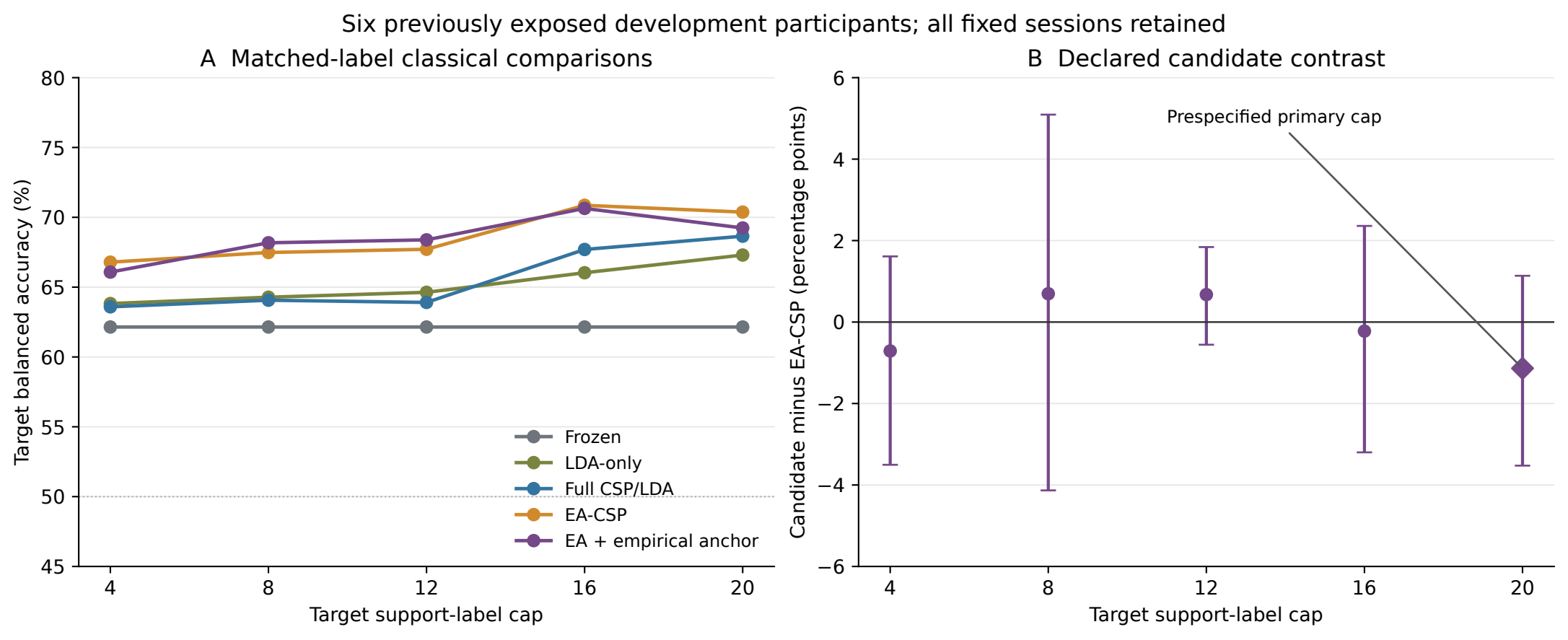
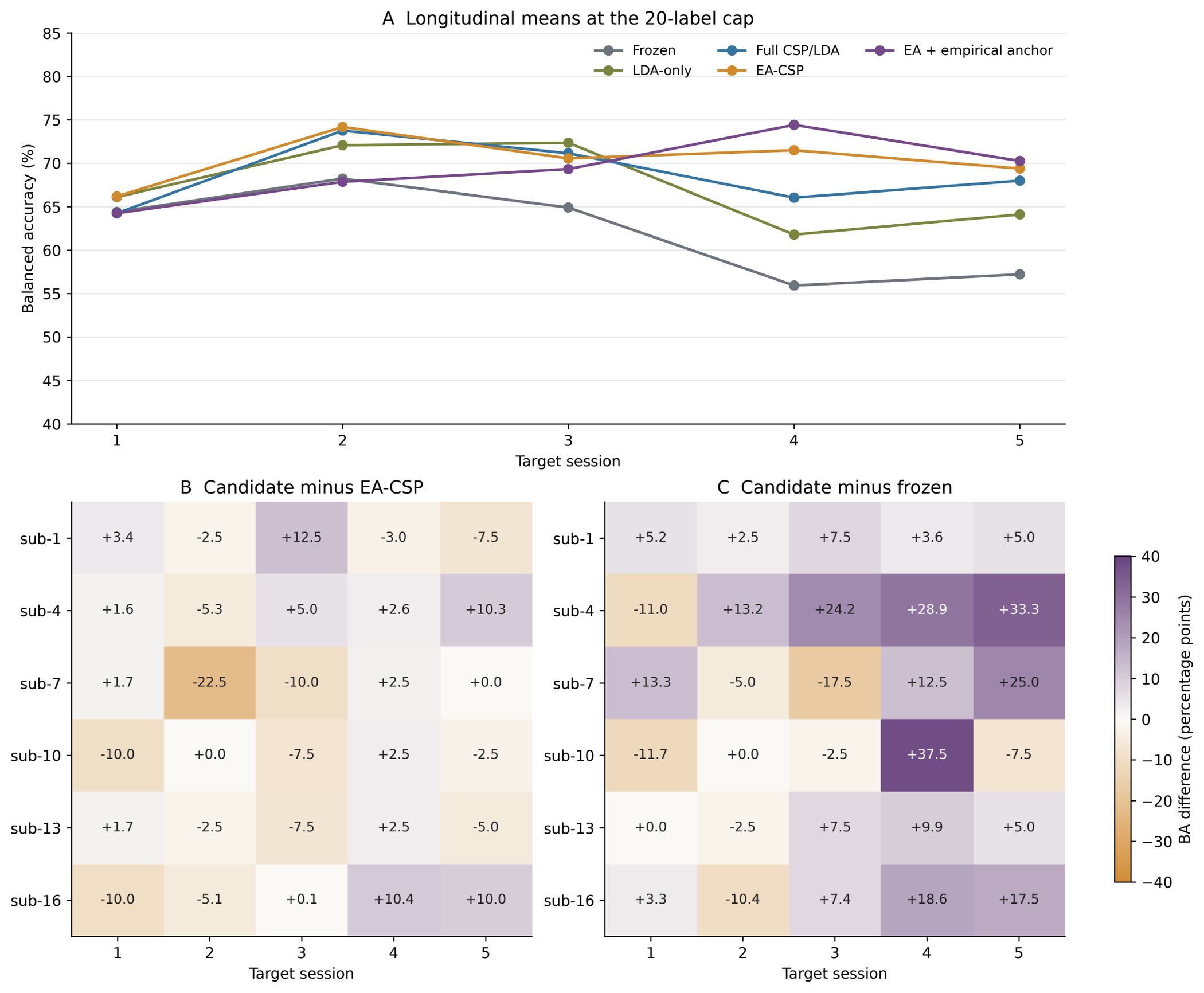
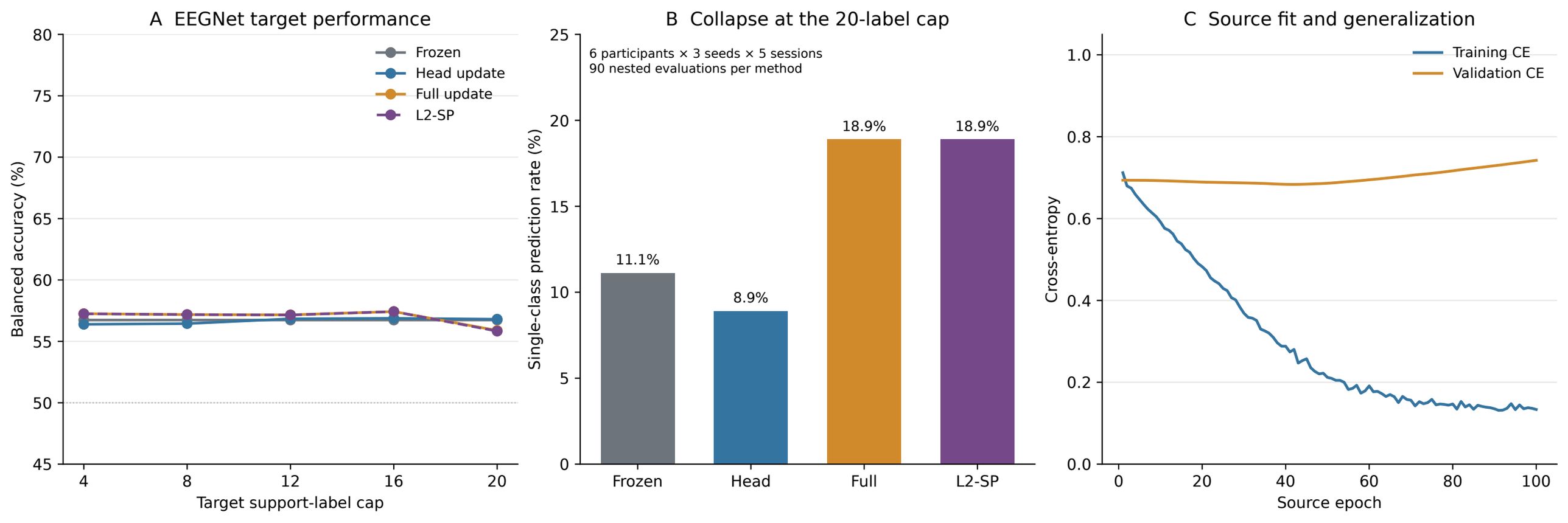
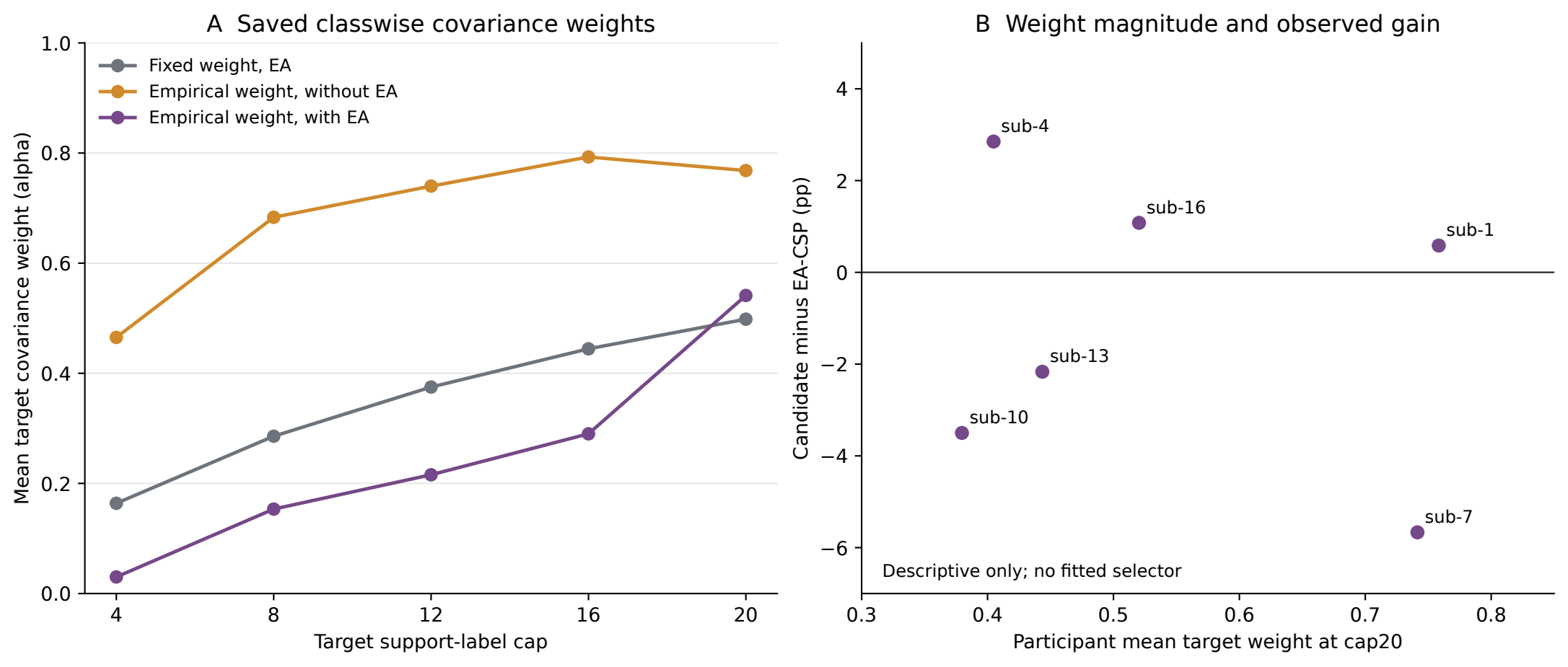

Six previously exposed development participants; all fixed sessions retained
A Matched-label classical comparisons
Target balanced accuracy (%)
80
75
70
65
60
55
50
45
4
8
12
16
20
Target support-label cap
Frozen
LDA-only
Full CSP/LDA
EA-CSP
EA + empirical anchor
B Declared candidate contrast
Candidate minus EA-CSP (percentage points)
6
4
2
0
−2
−4
−6
Prespecified primary cap
4
8
12
16
20
Target support-label cap


A Longitudinal means at the 20-label cap
Frozen
Full CSP/LDA
EA + empirical anchor
LDA-only
EA-CSP
Balanced accuracy (%)
Target session
B Candidate minus EA-CSP
sub-1 +3.4 -2.5 +12.5 -3.0 -7.5
sub-4 +1.6 -5.3 +5.0 +2.6 +10.3
sub-7 +1.7 -22.5 -10.0 +2.5 +0.0
sub-10 -10.0 +0.0 -7.5 +2.5 -2.5
sub-13 +1.7 -2.5 -7.5 +2.5 -5.0
sub-16 -10.0 -5.1 +0.1 +10.4 +10.0
C Candidate minus frozen
sub-1 +5.2 +2.5 +7.5 +3.6 +5.0
sub-4 -11.0 +13.2 +24.2 +28.9 +33.3
sub-7 +13.3 -5.0 -17.5 +12.5 +25.0
sub-10 -11.7 +0.0 -2.5 +37.5 -7.5
sub-13 +0.0 -2.5 +7.5 +9.9 +5.0
sub-16 +3.3 -10.4 +7.4 +18.6 +17.5
BA difference (percentage points)


A EEGNet target performance
Frozen
Head update
Full update
L2-SP
Balanced accuracy (%)
Target support-label cap
B Collapse at the 20-label cap
6 participants × 3 seeds × 5 sessions
90 nested evaluations per method
Single-class prediction rate (%)
11.1%
8.9%
18.9%
18.9%
Frozen
Head
Full
L2-SP
C Source fit and generalization
Training CE
Validation CE
Cross-entropy
Source epoch


A Saved classwise covariance weights
Fixed weight, EA
Empirical weight, without EA
Empirical weight, with EA
Mean target covariance weight (alpha)
1.0
0.8
0.6
0.4
0.2
0.0
4
8
12
16
20
Target support-label cap
B Weight magnitude and observed gain
Candidate minus EA-CSP (pp)
4
2
0
−2
−4
−6
sub-4
sub-16
sub-1
sub-13
sub-10
sub-7
Descriptive only; no fitted selector
0.3
0.4
0.5
0.6
0.7
0.8
Participant mean target weight at cap20